AI for Good Sandbox Hackathon – Ghayth Saleem – Muyassar (ميسّر)

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1. Introduction

   Muyassar is a multi-agent assistant that helps citizens in Jordan with government services and ties every answer to the article of a published law, policy or service guide it relies on. Four agents and an orchestrator cover the competition's roles: a procedure guide, a policy agent (legal advisor), an entity agent (institutional participant) and a document-check agent. Three failure scenarios expose where Jordanian law is silent, and each gap is compared with an EU, OECD or Estonian article the system retrieved.

2. Description of the use case

   For services such as passport, ID and birth registration (CSPD) and the disability card (HCD), citizens learn the steps, documents and fees from scattered pages, and rarely know which law governs a decision about them. When information arrives late, incomplete or to the wrong office, today's systems either decide automatically or leave the citizen without an explanation. Muyassar answers in plain Arabic (including Jordanian dialect), shows the exact article behind each statement, and stops before any sensitive decision until a staff member approves it.

   How it closes the gaps: every agent sees only the information delivered to it; a verdict with no retrieved article behind it is marked unsourced and shown in red; rejecting a request or keeping or sending health data stops the run for human approval; every step is a line in a JSONL audit log (time, agent, input, steps, articles, decision, confidence).

   Voice assistant and social inclusion: a citizen can speak the request in Jordanian Arabic, which serves older citizens and persons with disabilities. Speech is transcribed locally with faster-whisper (large-v3-turbo, CPU; a 12.1 s recording took 9.0 s), so the recording never leaves the machine and is never stored: only the transcript is kept, marked source: voice in the audit log. Scenario 2 turns on this: what the system heard becomes health data it must not keep without documented consent. Reading scanned documents (OCR) is future work; the prototype compares fields already extracted.

3. Use case requirement

   Knowledge base: 16 documents in three layers, 375 chunks. Laws are split at article boundaries, never by length, and every chunk carries doc_name, article, entity, date and url (enforced by a hook on kb/). Global documents also carry layer: global and jurisdiction.

Jordanian law: Personal Data Protection Law No. 24 of 2023 (secondary site, not yet matched to the Gazette); Law on the Rights of Persons with Disabilities No. 20 of 2017 (official PDF); Electronic Transactions Law No. 15 of 2015 (secondary site, not yet matched to the Gazette); Civil Status Law No. 9 of 2001, as amended (secondary site, not yet matched to the Gazette).

Jordanian policy: Jordan AI Strategy and Implementation Plan 2023–2027 (official PDF); National Charter of AI Ethics 2022 (official PDF); Digital Transformation Strategic Direction 2021–2025 (official PDF (English), Arabic translation ours).

Procedure guide: Instructions for the identification card for persons with disabilities, 2025 (agency website (Qistas copy)); CSPD service guide: passport renewal (official website); CSPD service guide: birth registration (official website); CSPD service guide: smart ID renewal (official website).

Global: EU AI Act 2024/1689 (selected articles) (EU Official Journal); OECD AI Principles (OECD/LEGAL/0449) (official website); Estonia X-Road (official website); GDPR arts 7, 9, 22 (EU Official Journal); SDG Regulation 2018/1724 arts 9, 10, 17 (EU Official Journal).

Retrieval: BM25 with Arabic normalisation plus multilingual-e5-large vectors in ChromaDB, fused by reciprocal rank fusion, then the top 30 reranked by bge-reranker-v2-m3. On 87 questions with known answers, written by the author: recall@5 = 0.977, recall@1 = 0.839.

Model: Claude Sonnet 5.5 with structured JSON output (a local Ollama model as fallback). Its sampling temperature cannot be fixed, so stability is measured instead (section 5).

Data protection: the API key stays in a git-ignored .env and the audit log is append-only, protected by a hook. A real pilot would need hosting and data residency agreed with the agency.

4. Mapping with ITU-T Y.3172, Readiness 2.0 Report, and domain-specific policies

Readiness 2.0: dimensions D1–D13 and their factors, from ITU's AI Ready Report 2.0 (January 2026).

Node	In Muyassar	Readiness 2.0 (dimension, factor)	Jordanian documents
SRC	Laws, policies, service guides; citizen requests and agency events	D1 Data/model Marketplace (Data, Open source)	PDPL 24/2023; Civil Status Law 9/2001; CSPD and HCD guides
C	Arabic PDF repair; source registry	D1 Data/model Marketplace (Data)	Electronic Transactions Law 15/2015 (electronic records)

PP	Article-boundary chunks with metadata; Arabic normalisation	D4 Contextualization and Regional Impact (Data, Research, Deployment)	Digital Transformation Strategic Direction 2021–2025
M	Hybrid retrieval + reranker; Claude, Gemini or a local model	D5 Level of Integration of AI in Workflows (Standards)	Jordan AI Strategy 2023–2027; National AI Ethics Charter 2022
P	Policy agent; citation and conflict rules; staff approval	D10 AI & Policies (Sandbox, Deployment); D9 Impacts of Humans in AI Integration	AI Ethics Charter (accountability, human oversight); PDPL arts 4, 5, 6
MLFO	Orchestrator and scenario engine	D8 Collaboration with AI (Open source, Standards)	Disability Rights Law 20/2017; HCD card instructions 2025
D / SINK	Web demo, voice input, replies, audit log	D6 Human Interface (Standards, Deployment); D11 AI for Inclusion	PDPL (data subject rights); Disability Rights Law art. 32
Sandbox	Local; logged runs replay without a key	Sandbox Environments (factor); D13 Digital Infrastructure (Deployment)	The ITU sandbox is optional and was not used.

Jordanian law, and a caveat: the Arabic of the PDPL 24/2023, the Electronic Transactions Law 15/2015 and the Civil Status Law 9/2001 comes from a secondary site (jordan-lawyer.com) and is not yet matched against the Official Gazette. The PDPL was cross-checked against MoDEE's official English translation, the ETL against MoDEE's PDF, and the Civil Status Law against the department's PDF (to 2015), which differs in articles 4, 30 and 32. A 2025 ETL amendment passed the House; its publication is unconfirmed. The Disability Rights Law is from an official PDF; its Gazette issue is unconfirmed.

5. Error scenarios

Each scenario is a JSON file of events, each with a simulated time and the one agent it reaches, so information arrives in batches, late and to different agents. Scenario 3's event

is hypothetical, and all personal data in the scenarios and the voice test recording is fictitious.

1. Doubtful document. The automatically read expiry date on an uploaded ID does not match the civil registry; two minutes later the registry check proposes automatic rejection. Who decides, what is the citizen told, and who answers for an automated error? Observed: the orchestrator suspends the rejection and refers the case to staff.

2. Health data in a disability-card request. The voice assistant hears a diagnosis and medication while a father applies; asked whether to keep them, he says «ماشي» (okay), and the Council asks for all medical details. Is spoken consent enough? Observed: PDPL art. 5 requires documented consent, so storage stops for approval. A run from a real recording took the same path, logged as source: voice.

3. Procedure changed suddenly (hypothetical event). The passport office changes the documents for minors before the knowledge base is updated, and a citizen asks. What does Muyassar say in between? Observed: the entity agent flags the contradiction with the published guide and the conflict rule holds the neutral interim reply for staff.

Stability: each scenario was run five times with identical input (logs/consistency/). 'Expected gap found' is a keyword test on the policy agent's gap text, defined in gap_check.

Scenario	Expected gap found	Stopped for staff approval (by the conflict rule)	Step without a source	Gap with retrieved global example
1. Doubtful document (+1 API error)	4/4	4/4 (0)	0/4	4/4
2. Health data in a disability-card request	5/5	5/5 (0)	0/5	5/5
3. Procedure changed suddenly	5/5	5/5 (5)	0/5	5/5

All three scenarios found their expected gap at least 4 times in 5. Scenario 1's fifth run stopped on an API credit error, so its row shows 4 completed runs and the error. Scenario 3 first stopped for approval in only 2 of 5 runs; a rule in code, not in the prompt, now makes any contradiction between an agency update and a retrieved published guide a sensitive decision, and it stopped 5 of 5.

Where the solution faces controversy. The staff stop trades speed for safety: a citizen waits, and Jordanian text sets no deadline for that review. In scenario 3 two agents disagree;

Muyassar does not pick a winner, it holds the reply and sends both views to staff. Gap detection is the model's reading of the texts, so each gap is shown with its articles for a person to check.

6. Policy gaps and standards gaps derived from the mapping exercise

   Each gap is what the policy agent reported, with the Jordanian articles it cited and a global-layer article it retrieved; a gap without one is shown in red.

Scenario	Gap the policy agent reported (latest run that passed the check)	Closest Jordanian text	Global example (retrieved)
1. Doubtful document	Automated rejection when a document does not match the registry (our translation; Arabic in reports/gaps.md)	Civil Status Law 9/2001 art. 4; Civil Status Law 9/2001 art. 2; PDPL 24/2023 art. 20; PDPL 24/2023 art. 5	GDPR art. 22
2. Health data in a disability-card request	No rule on who may receive a minor's health data, and none on whether spoken consent is enough (our translation; Arabic in reports/gaps.md)	PDPL 24/2023 art. 5; PDPL 24/2023 art. 6; PDPL 24/2023 art. 4; PDPL 24/2023 art. 10	GDPR art. 9
3. Procedure changed suddenly	No deadline for updating service information and no rule for the transition period (our translation; Arabic in reports/gaps.md)	DT Strategy 2021–2025; PDPL 24/2023 art. 4	SDG Regulation 2018/1724 art. 9

Update deadlines (scenario 3): the EU Single Digital Gateway Regulation 2018/1724 requires service information to be kept up to date and dated (art. 9) but sets no deadline in days. We found no official source that sets one, and cite none.

Standards gaps found while mapping. Readiness 2.0 names a factor for every dimension except D9 (humans in AI integration) and D11 (AI for inclusion), the two this solution leans on most. In our Y.3172 mapping, staff approval had to sit under the policy node and the audit log under the sink: we had no separate place for a step that holds an output for a person, or for a citation that travels from model to sink.

Links. Code, knowledge base, logs: github.com/ghayth-saleem/muyassar. Sources and verification: sources/registry.json. Gaps: reports/gaps.md. Stability: eval/consistency.md. Replay without a key: README.md.